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Graph Visualization in Python

November 9, 2023 Karmen Rabar

The power of graphs is already well known - graphs can represent complex data structures and relationships in various domains. According to different scenarios, for example, social networks, recommendation engines, or transportation systems, Python offers a range of graph data visualization

libraries, similar to the well-known NetworkX. In this blog post, we'll explore a few interesting methods and libraries for visualizing graphs in Python.

Pyvis

Pyvis is a Python library that simplifies the creation of interactive network graphs in a few lines of code. Pyvis is installed by running `pip install pyvis` in the command line. After that, you can import Pyvis and the `Network` module from Pyvis.

```
import pandas as pd
from pyvis.network import Network
pyvis.__version__
```

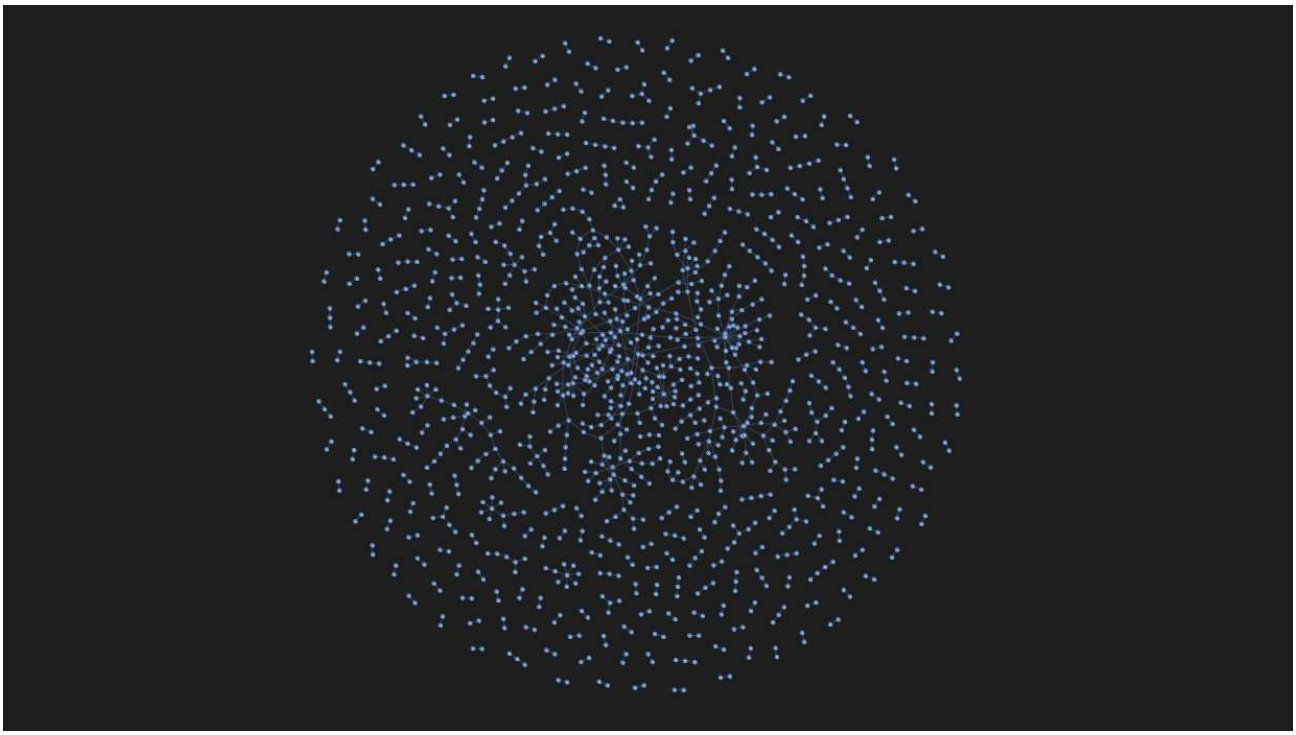
Visualizing social network

Let’s visualize an actual social network. We’ll use a dataset containing Facebook friend lists and limit our visualization to 1000 samples since this dataset is quite big with 88234 samples.

```
#Loading the data
data = pd.read_csv("facebook_combined.txt", sep=" ", header=None)
data.columns = ["person1", "person2"]
sample = data.sample(1000, random_state = 1)
sample.head(10)
```

To construct a network graph, start by creating an instance of the `Network` object. For visualization within a notebook, set `notebook` to `True` and choose `cdn_resources` to be `remote`. We'll set the nodes to be a unique list of people in the dataset and use the values of each row as source and destination nodes for each relationship. We’ll also use a dark background color and white font color.

```
net = Network(notebook = True, cdn_resources = "remote",
                bgcolor = "#222222",
                font_color = "white",
                height = "750px",
                width = "100%",
)
nodes = list(set([*sample.person1,*sample.person2]))
edges = sample.values.tolist()
net.add_nodes(nodes)
net.add_edges(edges)
net.show("graph.html")
```

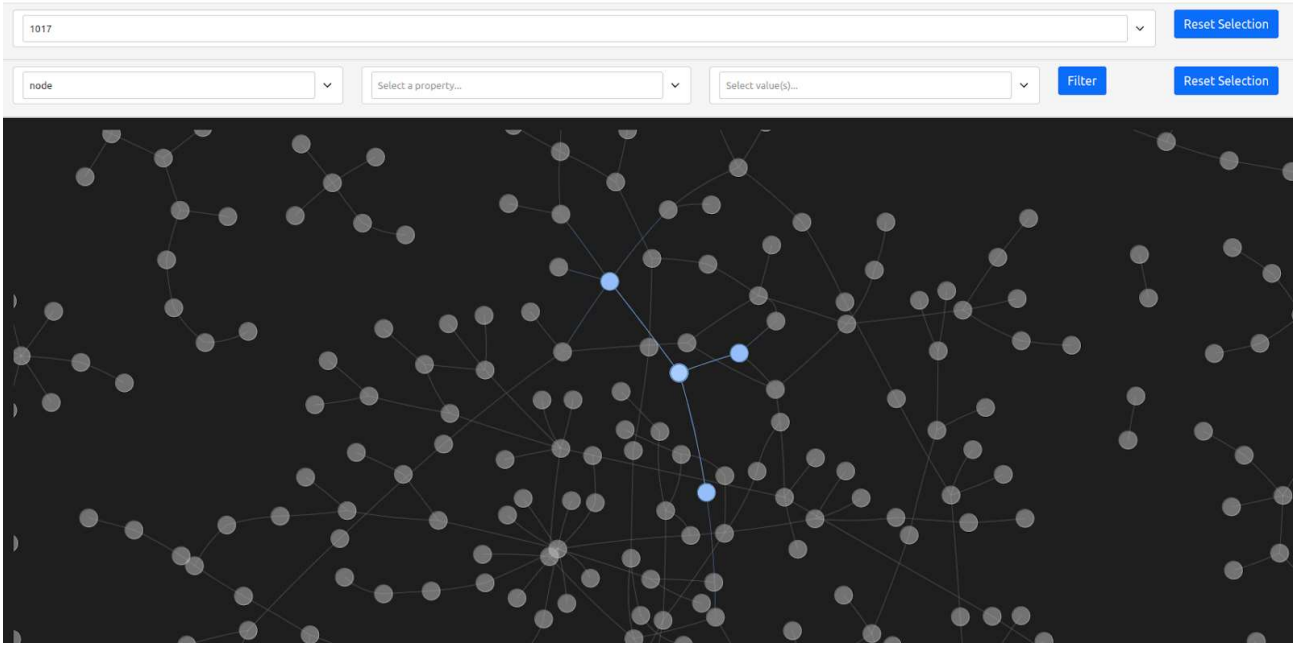


Filtering the results

With the graph now containing numerous nodes and relationships, we can filter and select nodes by adding the **`select_menu`** and **`filter_menu`** options as arguments. Now, we get a select bar and a filter bar. By specifying the ID of a node, we can select and highlight it, along with its adjacent edges and nodes.

```
net = Network(notebook = True, cdn_resources = "remote",
```

```
                    bgcolor = "#222222",
                    font_color = "white",
                    height = "750px",
                    width = "100%",
                    select_menu = True,
                    filter_menu = True,
)
nodes = list(set([*sample.person1,*sample.person2]))
edges = sample.values.tolist()
net.add_nodes(nodes)
net.add_edges(edges)
net.show("graph_with_menu.html")
```



Adjusting the graph's physics

The graph's physics can be adjusted using the **`show_buttons`** method with the **`filter_`** parameter set to **`physics`**. Altering parameters can yield intriguing visual effects. **`net.show_buttons(filter="physics")`**

After running the code, you will see the physics settings. For example, with the increase of the central gravity, nodes try to get to the middle because the gravity in the center of the graph increases, as shown in the picture above. If the spring length is increased, the distance between the nodes increases. I encourage you to explore various settings to achieve the desired visual impact. To replicate an effect, you can click `generate options`, copy the dictionary, and paste it into `net.set_options`.

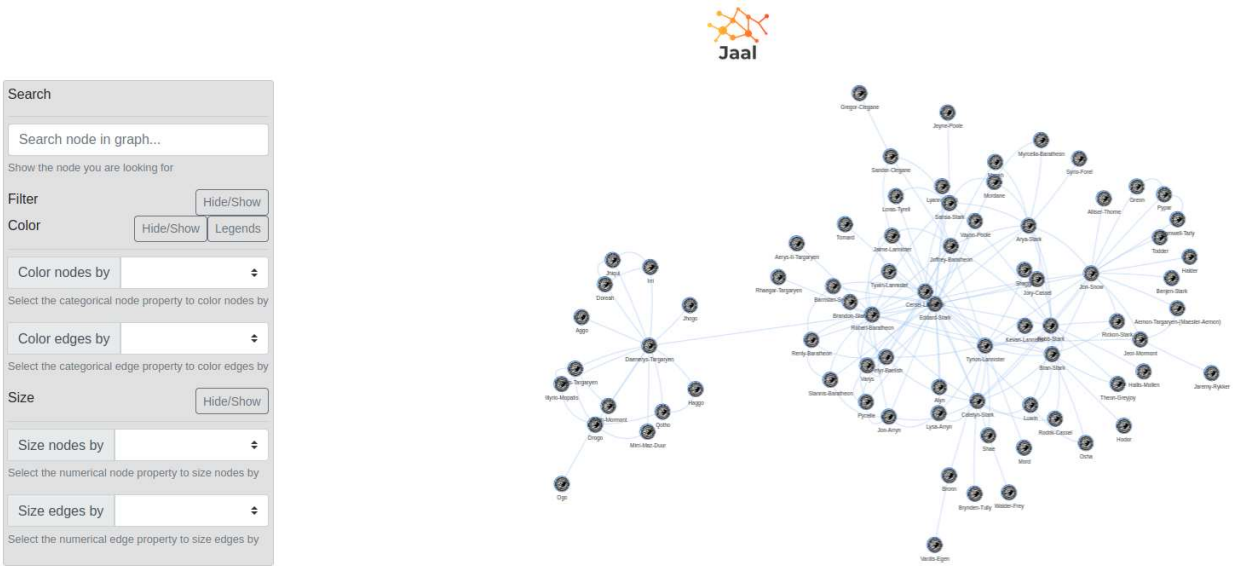
Jaal

Jaal is a Python-based interactive network visualization tool built by Mohit Mayank on the Dash and Visdcc libraries. You can install Jaal using the `pip install jaal` command. For best practices, it's recommended to create a virtual environment before installation. Once Jaal is installed, you can fetch

data and create visualizations. Let's explore this with the Game of Thrones dataset included in the package.

To visualize a network using Jaal, start by importing the Jaal main class and the dataset loading function, such as `load_got`. After that, load the Game of Thrones dataset using the provided function, resulting in two dataframes: `edge_df`, a pandas dataframe representing relationships, and `node_df`, an optional dataframe with unique node names. With the data loaded, pass it into Jaal and call the plot function. This action will launch Jaal on your localhost (127.0.0.1:8050), allowing you to interactively explore your network data. But, you can also use Jaal in your Jupiter notebook.

```
from jaal.datasets import load_got
#load the data
edge_df, node_df = load_got()
#init Jaal and run server
Jaal(edge_df, node_df).plot()
```

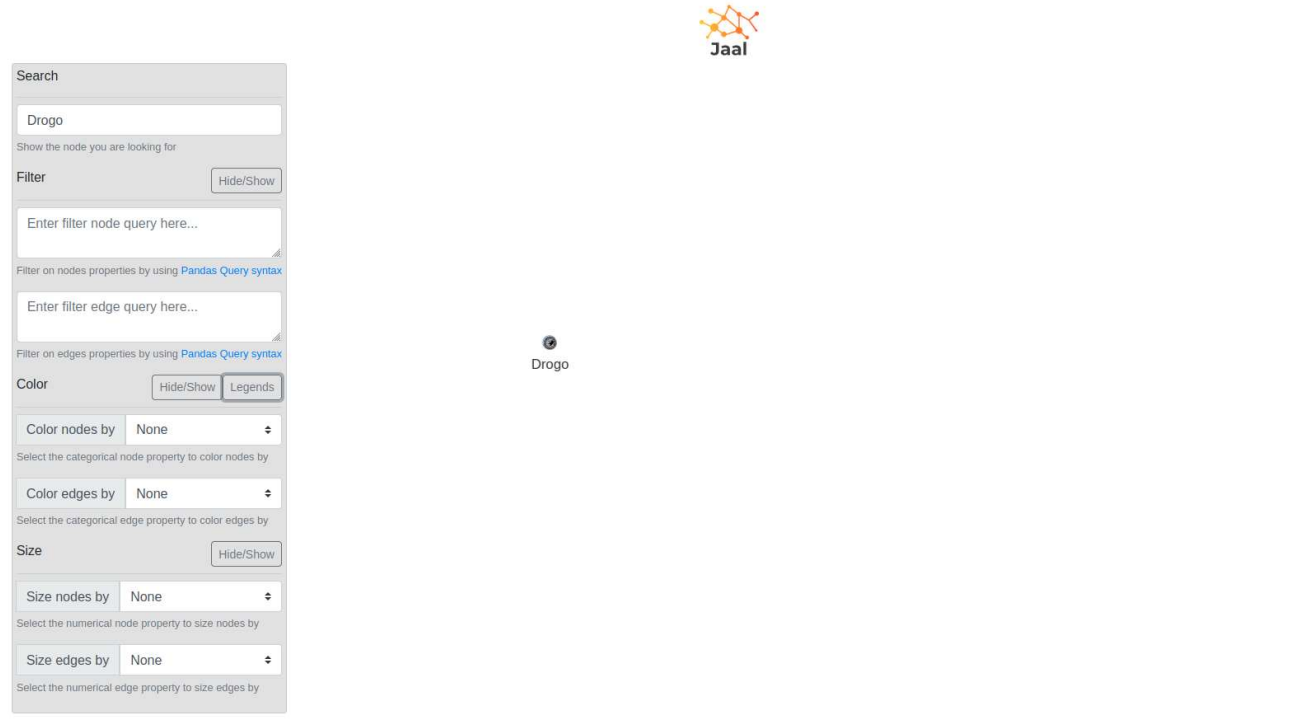


Jaal provides a dashboard with a settings panel on the left and a dynamic graph visualization on the right, making it a useful tool for network analysis.

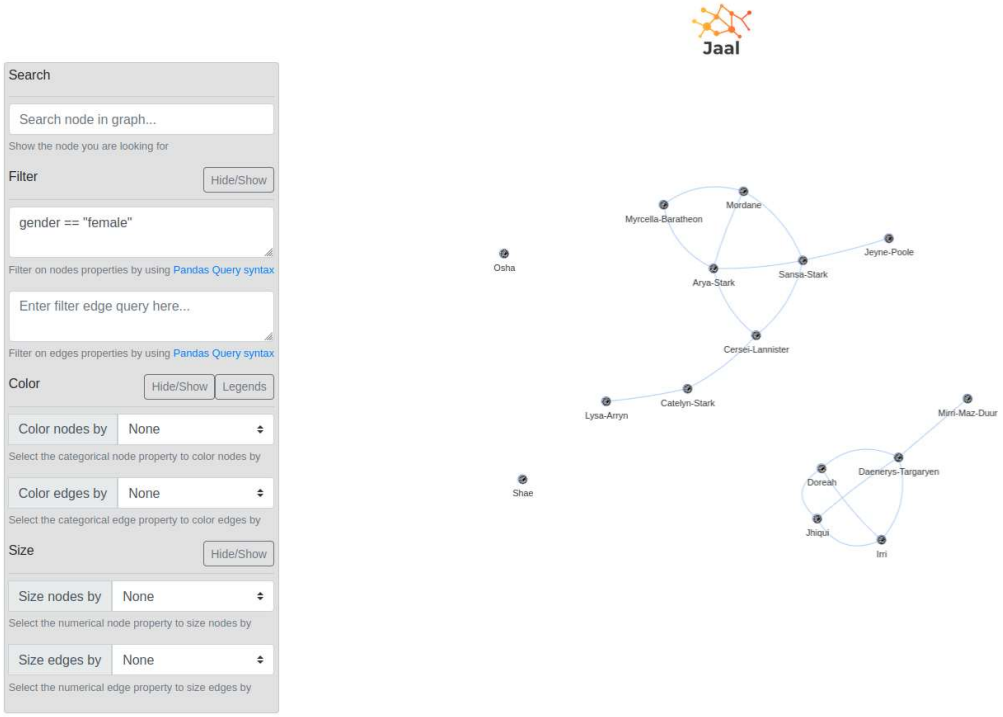
The Jaal dashboard consists of several sections, with the *Settings Panel* offering various options for graph data manipulation. Within this panel, you'll find tools such as *Search*, enabling character-by-character node searching, *Filter*, which supports filtering based on node or relationship features using Pandas query language, and *Color*, allowing nodes and relationships to be color-coded based on categorical features.

Searching, filtering and coloring

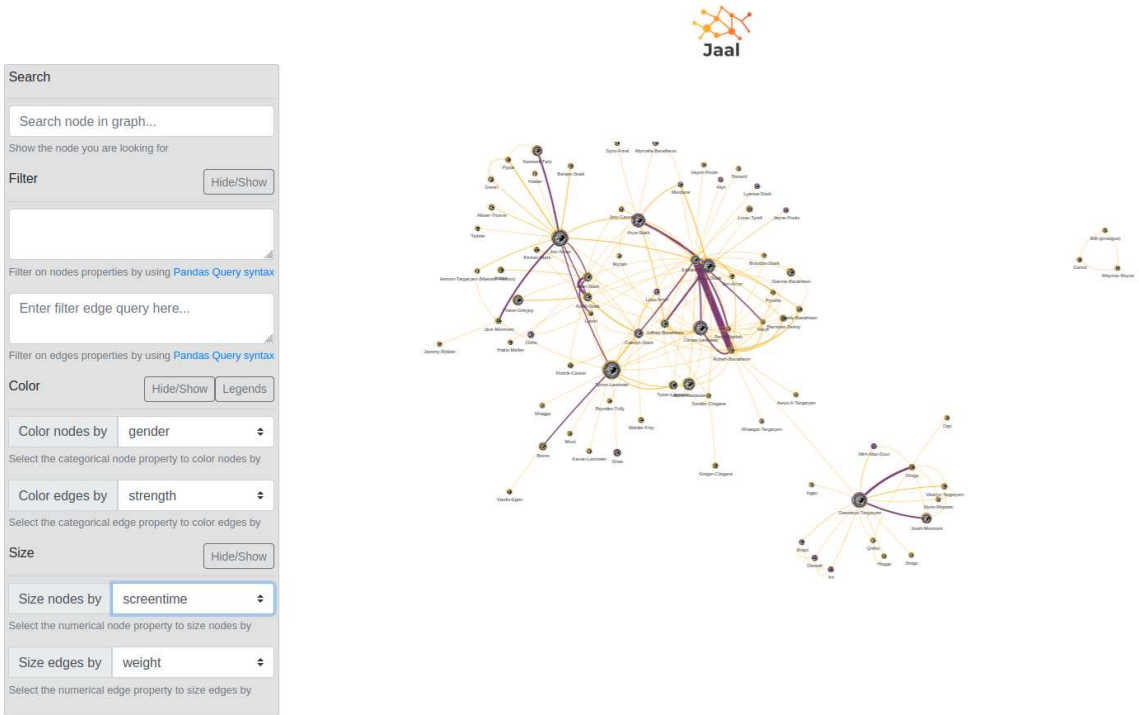
When it comes to searching, Jaal allows you to search for specific nodes within the graph, making it simple to find characters like **`Drogo`**.



Jaal supports filtering on both node and edge features, offering live previews of the filtering queries' effects. For instance, you can filter nodes based on criteria like **`gender == "female"`** and/or edges with conditions like **`weight > 20`**.



Coloring nodes and relationships is another valuable feature in Jaal. By using categorical features, you can visually represent feature distributions within the graph by color-coding elements, enhancing your understanding of the network's characteristics and relationships.



Takeaway

In conclusion, Pyvis provides a user-friendly way to create engaging network graphs with minimal coding effort. While Jaal is relatively new in Python's network visualization realm, it shows promise for future growth and collaboration. Whether you opt for Pyvis or Jaal, exploring these tools can unlock a wealth of valuable insights through effective graph visualization. Enjoy your journey into the world of graph visualization!